What happened when I opened a website

Open the example.com in browser and click enter

A dns query to resolve example.com

```
  ▸ [Timestamps]
    UDP payload (29 bytes)
▾ Domain Name System (query)
    Transaction ID: 0xfc5a
  ▸ Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  ▾ Queries
    ▾ example.com: type AAAA, class IN
        Name: example.com
        [Name Length: 11]
        [Label Count: 2]
        Type: AAAA (28) (IP6 Address)
        Class: IN (0x0001)
    [Response In: 934]
```

Server ip in response:

```
    Questions: 1
    Answer RRs: 2
    Authority RRs: 0
    Additional RRs: 0
  ▾ Queries
    ▾ example.com: type A, class IN
        Name: example.com
        [Name Length: 11]
        [Label Count: 2]
        Type: A (1) (Host Address)
        Class: IN (0x0001)
  ▾ Answers
    ▾ example.com: type A, class IN, addr 104.18.27.120
        Name: example.com
        Type: A (1) (Host Address)
        Class: IN (0x0001)
        Time to live: 90 (1 minute, 30 seconds)
        Data length: 4
        Address: 104.18.27.120
    ▾ example.com: type A, class IN, addr 104.18.26.120
        Name: example.com
        Type: A (1) (Host Address)
        Class: IN (0x0001)
        Time to live: 90 (1 minute, 30 seconds)
        Data length: 4
        Address: 104.18.26.120
    [Request In: 930]
    [Time: 8.443946 milliseconds]
```

TCP Packets:

Syn packet:

```
▸ Frame 935: Packet, 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface eth0, id 0
▸ Ethernet II, Src: PCSSystemtec_1f:b7:23 (08:00:27:1f:b7:23), Dst: NokiaShangha_c1:fa:50 (04:56:65:c1:fa:50)
▸ Internet Protocol Version 6, Src: 2401:4900:8814:fec0:51e9:dae9:3b40:1564, Dst: 2606:4700:90d2:8c7c:33a5:914
▾ Transmission Control Protocol, Src Port: 46800, Dst Port: 443, Seq: 0, Len: 0
    Source Port: 46800
    Destination Port: 443
    [Stream index: 20]
    [Stream Packet Number: 1]
  ▸ [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 3248869718
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 0
    Acknowledgment number (raw): 0
    1010 .... = Header Length: 40 bytes (10)
  ▸ Flags: 0x002 (SYN)
    Window: 64800
    [Calculated window size: 64800]
    Checksum: 0xa759 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  ▸ Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale
  ▸ [Timestamps]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
```

Syn, Ack:

```
942 37.162699287  2606:4700:90d2:8c7c  2401:4900:8814:fec0  TLSv1.3  1304 Server Hello
▸ Internet Protocol Version 6, Src: 2606:4700:90d2:8c7c:33a5:914:ccc4:a209, Dst: 2401:4900:8814:fec0:51e9:dae
▾ Transmission Control Protocol, Src Port: 443, Dst Port: 46800, Seq: 0, Ack: 1, Len: 0
    Source Port: 443
    Destination Port: 46800
    [Stream index: 20]
    [Stream Packet Number: 2]
  ▸ [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 3525226733
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 3248869719
    1010 .... = Header Length: 40 bytes (10)
  ▸ Flags: 0x012 (SYN, ACK)
    Window: 65535
    [Calculated window size: 65535]
    Checksum: 0x743b [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  ▸ Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale
  ▸ [Timestamps]
  ▾ [SEQ/ACK analysis]
      [This is an ACK to the segment in frame: 935]
      [The RTT to ACK the segment was: 28.707574 milliseconds]
      [iRTT: 28.854125 milliseconds]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
```

Ack packet:

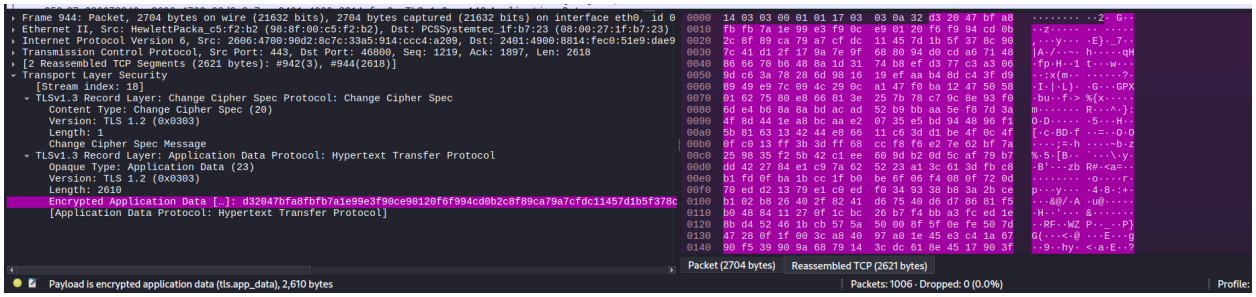
```
935 37.162723501  2401:4900:8814:fec0  2606:4700:90d2:8c7c  TCP  94 46800  443 [SYN] Seq=0 Win=6480
▸ Frame 938: Packet, 1434 bytes on wire (11472 bits), 1434 bytes captured (11472 bits) on interface eth0, id 0
▸ Ethernet II, Src: PCSSystemtec_1f:b7:23 (08:00:27:1f:b7:23), Dst: NokiaShangha_c1:fa:50 (04:56:65:c1:fa:50)
▸ Internet Protocol Version 6, Src: 2401:4900:8814:fec0:51e9:dae9:3b40:1564, Dst: 2606:4700:90d2:8c7c:33a5:914
▾ Transmission Control Protocol, Src Port: 46800, Dst Port: 443, Seq: 1, Ack: 1, Len: 1348
    Source Port: 46800
    Destination Port: 443
    [Stream index: 20]
    [Stream Packet Number: 4]
  ▸ [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 1348]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 3248869719
    [Next Sequence Number: 1349    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 3525226734
    1000 .... = Header Length: 32 bytes (8)
  ▸ Flags: 0x010 (ACK)
    Window: 254
    [Calculated window size: 65024]
    [Window size scaling factor: 256]
    Checksum: 0xac95 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  ▸ Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  ▸ [Timestamps]
  ▾ [SEQ/ACK analysis]
      [iRTT: 28.854125 milliseconds]
      [Bytes in flight: 1348]
      [Bytes sent since last PSH flag: 1348]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (1348 bytes)
    [Reassembled PDU in frame: 939]
    TCP segment data (1348 bytes)
```

TLS Connection:

Client Hello:

```
No.   Time          Source            Destination       Protocol  Length  Info
933 37.100682956  192.168.1.1       192.168.1.2       DNS       103     Standard query response 0x8f5d
▸ Ethernet II, Src: PCSSystemtec_1f:b7:23 (08:00:27:1f:b7:23), Dst: NokiaShangha_c1:fa:50 (04:56:65:c1:fa:50)
▸ Internet Protocol Version 6, Src: 2401:4900:8814:fec0:51e9:dae9:3b40:1564, Dst: 2606:4700:90d2:8c7c:33a5:91
▸ Transmission Control Protocol, Src Port: 46800, Dst Port: 443, Seq: 1349, Ack: 1, Len: 548
▸ [2 Reassembled TCP Segments (1896 bytes): #938(1348), #939(548)]
▾ Transport Layer Security
    [Stream index: 18]
  ▾ TLSv1.3 Record Layer: Handshake Protocol: Client Hello
      Content Type: Handshake (22)
      Version: TLS 1.0 (0x0301)
      Length: 1891
    ▾ Handshake Protocol: Client Hello
        Handshake Type: Client Hello (1)
        Length: 1887
      ▸ Version: TLS 1.2 (0x0303)
        Random: 82e7424fa4ccc92fcd07291d194be6735dd12efc1deb274104b208b08a548ac6
        Session ID Length: 32
        Session ID: dcc30554f8c48dbc8990b9d6ef3a2093f0c1efeb08f31a5b5f1119c953a731f7
        Cipher Suites Length: 34
      ▸ Cipher Suites (17 suites)
        Compression Methods Length: 1
      ▸ Compression Methods (1 method)
        Extensions Length: 1780
      ▸ Extension: server_name (len=16) name=example.com
      ▸ Extension: extended_master_secret (len=0)
      ▸ Extension: renegotiation_info (len=1)
      ▸ Extension: supported_groups (len=16)
      ▸ Extension: ec_point_formats (len=2)
      ▸ Extension: session_ticket (len=0)
      ▸ Extension: application_layer_protocol_negotiation (len=14)
      ▸ Extension: status_request (len=5)
      ▸ Extension: delegated_credentials (len=10)
      ▸ Extension: signed_certificate_timestamp (len=0)
      ▸ Extension: key_share (len=1327) X25519MLKEM768, x25519, secp256r1
      ▸ Extension: supported_versions (len=5) TLS 1.3, TLS 1.2
      ▸ Extension: signature_algorithms (len=24)
      ▸ Extension: psk_key_exchange_modes (len=2)
      ▸ Extension: record_size_limit (len=2)
      ▸ Extension: compress_certificate (len=7)
      ▸ Extension: encrypted_client_hello (len=281)
        [JA4: t13d1717h2_5b57614c22b0_3cbfd9057e0d]
        [JA4_r: t13d1717h2_002f,0035,009c,009d,1301,1302,1303,c009,c00a,c013,c014,c02b,c02c,c02f,c030,cca8,c
        [JA3 Fullstring: 771,4865-4867-4866-49195-49199-52393-52392-49196-49200-49162-49161-49171-49172-156-
        [JA3: 6f7889b9fb1a62a9577e685c1fcfa919]
```

Server Hello:

```
▸ Frame 942: Packet, 1304 bytes on wire (10432 bits), 1304 bytes captured (10432 bits) on interface eth0, id 0
▸ Ethernet II, Src: NokiaShangha_c1:fa:50 (04:56:65:c1:fa:50), Dst: PCSSystemtec_1f:b7:23 (08:00:27:1f:b7:23)
▸ Internet Protocol Version 6, Src: 2606:4700:90d2:8c7c:33a5:914:ccc4:a209, Dst: 2401:4900:8814:fec0:51e9:dae9
▸ Transmission Control Protocol, Src Port: 443, Dst Port: 46800, Seq: 1, Ack: 1897, Len: 1218
▾ Transport Layer Security
    [Stream index: 18]
  ▾ TLSv1.3 Record Layer: Handshake Protocol: Server Hello
      Content Type: Handshake (22)
      Version: TLS 1.2 (0x0303)
      Length: 1210
    ▾ Handshake Protocol: Server Hello
        Handshake Type: Server Hello (2)
        Length: 1206
      ▸ Version: TLS 1.2 (0x0303)
        Random: c5122e15f11a7898ffb5016f10cc872194c9e351b8b965a7af422cb8800d0fe8
        Session ID Length: 32
        Session ID: dcc30554f8c48dbc8990b9d6ef3a2093f0c1efeb08f31a5b5f1119c953a731f7
        Cipher Suite: TLS_AES_128_GCM_SHA256 (0x1301)
        Compression Method: null (0)
        Extensions Length: 1134
      ▸ Extension: key_share (len=1124) X25519MLKEM768
      ▸ Extension: supported_versions (len=2) TLS 1.3
        [JA3S Fullstring: 771,4865,51-43]
        [JA3S: eb1d94daa7e0344597e756a1fb6e7054]
```

Cipher spec change with data encrypted: